

AI Autonomy Driving a Practice-Oriented Logistics Paradigm Shift

Preview

Checking dozens of emails pouring in every day, juggling complex manual Excel work across multiple systems and documents, and dealing with unexpected global port delays.

Frontline workers in logistics are struggling amid fragmented communication and endless repetitive tasks. While the first wave of digital transformation (1st Wave) based on platforms has significantly improved visibility compared to past analog methods, there is still a wide area where people must manually check data and handle exceptions.

This white paper goes beyond the simple introduction of IT systems to shed light on the intelligent logistics innovation (2nd Wave) that Large Language Models (LLM) and intelligent agents (AI Agents) enable by directly understanding the work context of working-level workers and supporting genuine “hyper-automation.” Based on actual operational cases and the technology roadmap of Samsung SDS Cello Square, it provides a detailed analysis of how everyday operational processes—from quotation, shipping, and visibility/risk management to logistics cost settlement—are fundamentally changing. Through this, it aims to present the new roles and values that working-level workers should prepare for in the upcoming intelligent work environment.



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01.

Prologue: The Trend of Digital Transformation in Logistics and the Settlement of the 1st Wave

PART 1. Prologue: The Trend of Digital Transformation in Logistics and the Settlement of the 1st Wave

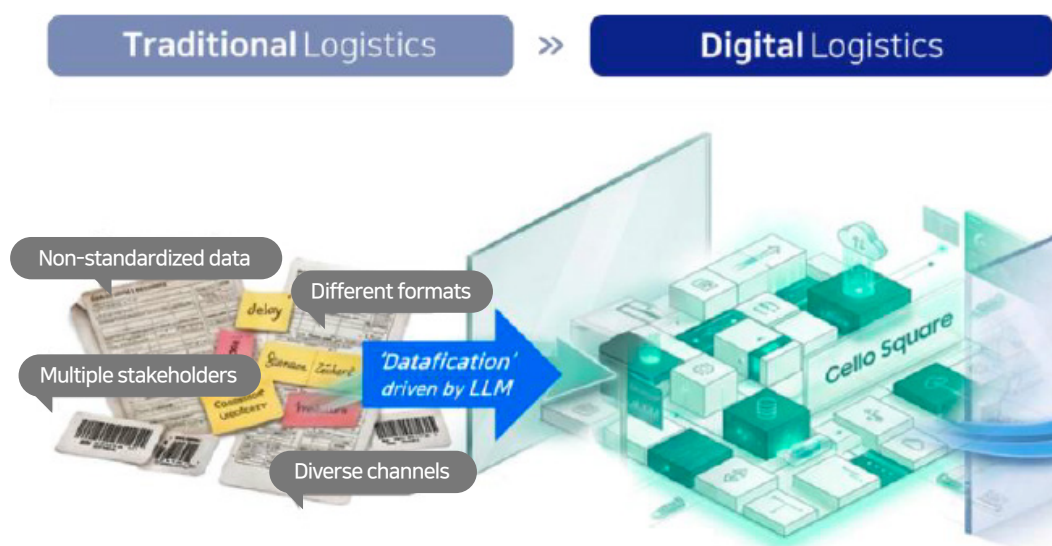
Just a few years ago, export and import logistics operations were a series of analog manual processes that relied entirely on phone calls and emails. Transporting a single cargo required up to 10 to 20 types of document processing, and communication among numerous stakeholders such as forwarders, shipping lines, shippers, and customs was conducted through different formats and fragmented channels. To check the real-time location of cargo, the person in charge had to access each shipping line's website to manually look up the B/L number or had to send emails to partner companies. These information silos and passive methods were the main causes of missing the golden time for responding to exceptions such as port skips or customs clearance delays, and they significantly increased overtime work and work fatigue for operators.

The emergence of platform-based digital logistics (Digital Logistics, 1st Wave) is what has appeared to overcome these limitations and secure visibility. The introduction of integrated digital logistics platforms, such as Samsung SDS's Cello Square, has brought about a revolutionary change in logistics operations. The biggest changes were the "centralization of data" and the "integration of business processes."

In the digital logistics environment, operators no longer need to gather scattered Excel files here and there. When receiving quotes for freight transportation, instead of negotiating over the phone, they can simply access a platform, enter the origin and destination, and instantly compare various transportation options and online quotes. The overwhelming shipment documents are systematically uploaded within the platform, making them searchable and accessible at any time, and through system integration, essential documents such as bills of lading (B/L) and commercial invoices (CI) are generated using standardized templates.

In particular, progress in terms of "securing visibility" has been remarkable. In the past, it was difficult to know whether cargo was in the middle of the sea or had arrived at the port, but in the 1st Wave environment, it has become possible to track the movement routes of cargo in real-time on a global map based on an API network. Through this, even when inquiries such as "When will our cargo arrive?" come from other internal departments (sales, production, etc.), operators can open the dashboard and share the exact estimated arrival date with just a few clicks. In this way, the first digital transformation played a key role in laying the foundation for a transparent and efficient logistics ecosystem by systematically accumulating standardized data within the system.

[Figure] 1st Wave: Digital Transformation of Traditional Logistics



02.

Body: Innovation in Logistics Practices with AI Agents(2nd Wave)

AI Agent

PART 2. Body: Innovation in Logistics Practices with AI Agents (2nd Wave)

Despite the fact that building digital platforms has made work much more convenient, the subject who still inputs information into the system is “human.” Unstructured data, such as special requests exchanged by email or agreements on exceptional costs, is not easily captured in the system. In an environment where AI reads and datafies unstructured data, groundbreaking features that go beyond simple monitoring and assist operators in making decisions are being introduced. Let's take a detailed look at what changes are taking place at each key stage of international transportation operations through the example of Samsung SDS Cello Square.

1. Quoting and Sourcing: Multidimensional Analysis Calculating Even Hidden Risks

Beyond logging onto the platform and getting an instant quote, skilled on-site workers do not blindly choose just the “cheapest option on the rate sheet.” Drawing on accumulated experience, they consider hidden risks such as “this carrier tends to have long transshipment times at certain ports” or “delays are frequent during this season due to weather issues.”

AI agents replace the intuition of these practical experts with data analysis. They provide real data such as how well each carrier keeps its ETA and whether there have been delays in lead times on specific routes. In addition, they present expected additional costs and carbon emissions (ESG indicators) by transportation method. Operators can easily select the optimal routing options analyzed by AI according to urgency or cost reduction goals.

2. Booking and Shipping: 24/7 Email-based Automation Agent

The biggest pain point in booking operations is manually checking and entering the information from the shipment documents (B/L, invoice, packing list, etc.) that pour in via email. When a booking email comes in during the night or on weekends after the operators have left work, the system remains in a “blank” state until they return to the office the next day.

The AI-based Booking Agent operates 24 hours a day. As soon as an email arrives, it identifies the context of which Purchase Order (PO) it receives through natural language processing capabilities and automatically registers it in the system. In particular, if there is a discrepancy in the documents, such as a mismatch between the total weight of goods on the commercial invoice and the packing list, it cross-checks past master data and immediately sends a verification alert to the relevant personnel. The responsible person transitions from simple typing tasks to a managerial role, approving only the abnormal cases.

3. Visibility Management: Control Network That Fills in Missing Tracking Information

The visibility of digital platforms also experiences cases where some location information is omitted due to system integration issues or carrier circumstances. In such cases, previously, operators had to manually check carrier websites or make phone calls to fill in the blanks.

Samsung SDS's AI system learns from a six-figure number of tracks through machine learning to calculate not only the predictive ETA¹ but also the predictive ETD. When it detects that essential information is missing from the system, the AI agent crawls carriers' websites or directly sends an email or messenger to the partner company. It then reads the response and automatically fills in the blanks in the system, completing an unmanned control loop.

4. Risk Management: From News Sensing to Item-Level Impact

In the event of a canal blockage or a local port strike, existing operators had to manually filter through Excel to see how much of their company's cargo was involved in the route. AI risk management senses 60,000 pieces of global news daily, refining and informing only the real risks that affect the company.

Furthermore, by combining the document information read by AI during the past booking stage, it analyzes the business impact at the item level and reports not just "the container is delayed," but "the essential production line component (PO number 123) loaded in that container is delayed."

¹ Predictive ETA: A recalculated arrival time based on real-time data and AI analysis, rather than past schedule-based estimates

5. Logistics Cost Settlement (Billing/Settlement): Meticulousness in Identifying Hidden Extra Costs in Emails

The settlement work to check if the contracted freight rates match the invoiced amounts is extremely exhausting. This is because countless “exceptional costs,” such as terminal delay charges or fuel surcharges, which are negotiated via email, get mixed in.

The AI settlement engine automatically matches the base rates and invoices from contracts item by item, and immediately approves transactions where the amounts match. Additionally, it scans non-standard details previously agreed upon in past emails, filters out only the surcharges with issues as “exceptions/return targets”, and requests a review from the relevant personnel, thereby completely blocking cost leakage.

[Table] The Evolution of Logistics Operations Paradigm: From Traditional Logistics to Intelligent Logistics

Work area	Traditional Logistics	Digital Logistics (1 st Wave)	AI-Powered Logistics (2 nd Wave)
Quotation/ Sourcing	Manual-based unit rate comparison, information consolidation takes several days	Real-time online quotation inquiry on the platform and Intuitive rate comparison	Comprehensive analysis and optimal alternative route recommendation with past delay history, port situation, carbon emissions, etc.
Booking/ Shipping	After receiving the documents via emails, manual system typing, frequent human errors	Direct order registration within the platform and integrated management of standard template-based documents	After understanding the context of the received email, automatic booking within 24 hours and cross-verification
Visibility Management	Carrier website inquiries, cargo location tracking disruptions, and time differences	API Integration for map-based global cargo real-time tracking and monitoring	Automatic Prediction of Departure/Arrival (ETA/ETD), automatic crawling and self-complementary update upon detecting missing information
Risk Management	When a global issue occurs, manually filter company's cargo impact	Visualization of major delay notifications and exception situation reports within the platform dashboard	After real-time global new sensing and impact analysis, send notification business impact by “Item (PO) unit”
Logistics cost settlement	Manual transaction of contract freights and invoices, frequent omissions	Integrated search of settlement history based on platform-registered transportation performance history and costs	Bill-contract auto-matching, approval/rejection of agreed exception costs detected in emails

03.

Epilogue: The Integration of Practice and AI for True Autonomy

PART 3. Epilogue: The Integration of Practice and AI for True Autonomy

Decision-making Reasons That System Doesn't Know, AI That Understands "Why"

Likewise, although advanced AI technology is being deployed on site, it is difficult for all logistics operations to become fully unmanned in a short period of time. This is because existing structured systems only record what results the person in charge has derived (What), not why such a decision was made (Why).

In real-world business scenarios, there are numerous cases where companies choose suppliers with higher rates due to "context", such as delivery accident histories or strong demands from clients, even though certain suppliers offer lower rates. The future AI agents, specifically the "Knowledge Graph" and "Context Layer" built on top of Cello, the foundational system of Samsung SDS, are designed to address this very issue. The AI learns from the fragmented exception handling methods and decision histories (Decision Memory) found in messengers, meeting minutes, and emails, enabling it to make rational judgments at the same level as humans.

The New Positioning of Operators Preparing for the Future

As AI becomes deeply integrated into the logistics field, the roles of operators must also undergo a revolutionary transformation. Repetitive tasks such as entering invoices every morning, refreshing carrier websites to check cargo locations, and reconciling Excel sheets at the end of each month should gradually be fully delegated to AI agents. Since trends in the field are changing so rapidly, it is important to smartly leverage the infrastructure of already proven digital logistics platforms like Cello Square, rather than struggling to integrate fragmented IT tools directly.

Going forward, the core competency required of operators is not the speed of manual processing. It is the insight as a "decision-maker" who controls supply chain exceptions based on multidimensional data generated by AI, plans rapid alternative sourcing routes, and minimizes cross-departmental business impact. In the era beyond digital transformation, where AI works on its own, logistics managers will break free from the cycle of repetitive tasks and establish themselves as key players at the forefront of corporate management, maximizing supply chain value.



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